

OUTPUT

1. A* SEARCH ALGORITHM

```
=====
A* SEARCH DRY RUN
=====
```

```
Iteration 1
```

```
-----
Current Node : A
```

```
Open List:
  Empty
```

```
Closed List:
  Empty
```

```
Current Node Values
g(A) = 0
h(A) = 7
f(A) = 7
```

```
Iteration 2
```

```
-----
Current Node : B
```

```
Open List:
  C (f=8)
```

```
Closed List:
  ['A']
```

```
Current Node Values
g(B) = 2
h(B) = 6
f(B) = 8
```

```
Iteration 3
```

```
-----
Current Node : E
```

```
Open List:
  C (f=8)
  D (f=12)
```

```
Closed List:
```

['A', 'B']

Current Node Values

$g(E) = 4$

$h(E) = 2$

$f(E) = 6$

Iteration 4

Current Node : C

Open List:

D ($f=12$)

Closed List:

['A', 'B', 'E']

Current Node Values

$g(C) = 4$

$h(C) = 4$

$f(C) = 8$

Iteration 5

Current Node : D

Open List:

Empty

Closed List:

['A', 'B', 'E', 'C']

Current Node Values

$g(D) = 9$

$h(D) = 3$

$f(D) = 12$

Iteration 6

Current Node : G

Open List:

Empty

Closed List:

['A', 'B', 'E', 'C', 'D']

Current Node Values

$g(G) = 11$

$h(G) = 0$

$f(G) = 11$

=====

FINAL RESULT

=====

Optimal Path : A -> B -> D -> G

Total Path Cost : 11

2.MINIMAX WITH ALPHA-BETA PRUNING

===== Minimax with Alpha-Beta Pruning =====

Step 1

Current Node : MAX

Alpha = -inf

Beta = inf

Move to Left MIN

Step 2

Current Node : Left MIN

Visited Node : 3

Alpha = -inf

Beta = 3

Selected Value = 3

Step 3

Current Node : Left MIN

Visited Node : 5

Alpha = -inf

Beta = 3

Selected Value = 3

Step 4

Current Node : Left MIN

Visited Node : 6

Alpha = -inf

Beta = 3

Selected Value = 3

Left MIN returns 3

Step 5

Back to MAX

Alpha = 3

Beta = inf

Current Best = 3

Move to Right MIN

Step 6

Current Node : Right MIN

Visited Node : 9

Alpha = 3

```
Beta = 9
Selected Value = 9
-----
Step 7
Current Node : Right MIN
Visited Node : 1
Alpha = 3
Beta = 1
Selected Value = 1
Alpha-Beta Cutoff Occurs
-----
Step 8
Pruned Node : 2
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===== FINAL RESULT =====
Left MIN Value  = 3
Right MIN Value = 1
Final Minimax Value = 3
Best Move for MAX : Left Subtree
Pruned Node : 2
|
```